

CMPE 413

Fall 2025

Lab Final Project Extra Credit:
2-Way Cache VHDL Code and Layout

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Section 1 Mon/Wed 1:00 PM–2:15 PM

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2-Way Cache Diagrams

2.1 Block Diagram of Hierarchical Design

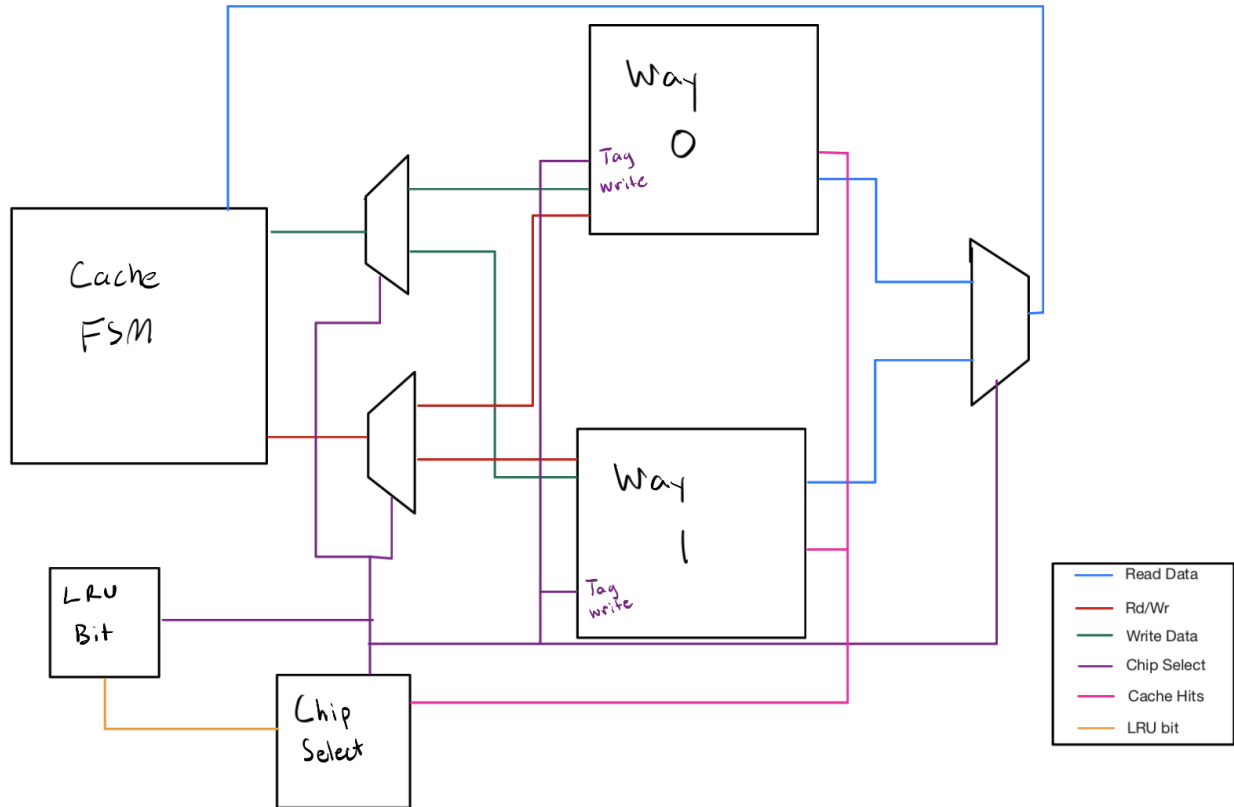


Figure 1: 2-Way Cache Design

Figure 1 shows the full design of the 2-way cache. There is only one FSM that controls both ways. The inputs and outputs for this cache and several signals are the same as the 1-Way, only the signals that are changed are shown in the diagram. The cache sends a controlled read write signal and write data signal to the way that is enabled. The address is always sent to both ways to determine hits, which are used to decide which way to send written data to. The read data from each way during a read operation is demuxed so only the chosen way has its data outputted. The chip select uses the LRU bit and the cache hits to determine which way should be used.

2.2 Chip Selection

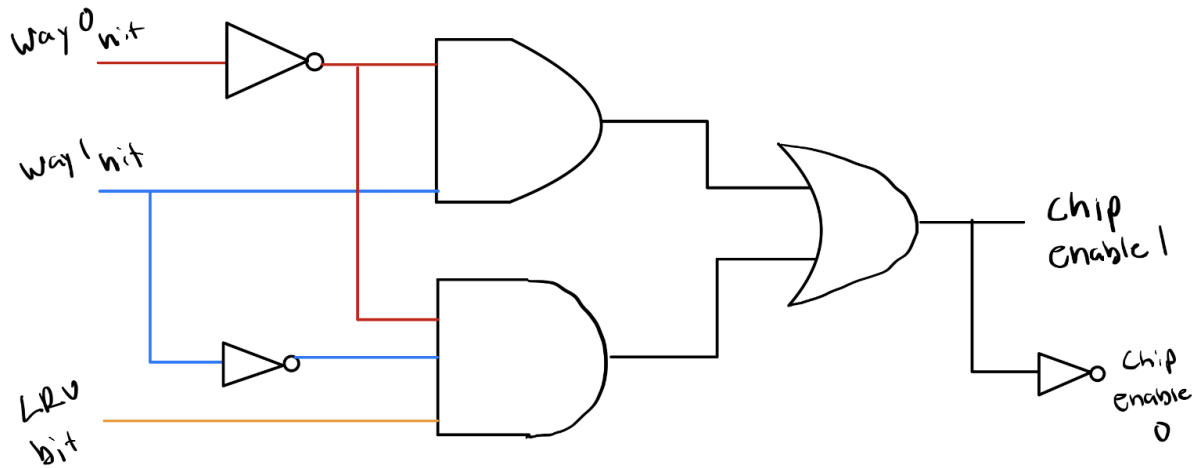


Figure 2: Chip Selection Diagram

Figure 2 shows the design idea of the chip select. This design determines whether chip 1 should be enabled, chip 0 enable is always the inverse of chip 1 enable. Chip 1 enable only goes high if way 0 was a miss and way 1 was a hit, or both ways miss and the LRU bit is high. A high LRU bit represents way 1 and low LRU bit represents way 0. The chip enable is used for the selects for the muxes and demuxes so data can flow to and from the correct ways. The enable is also used to enable whether a tag should be written in a read miss operation.

2.3 Updating LRU

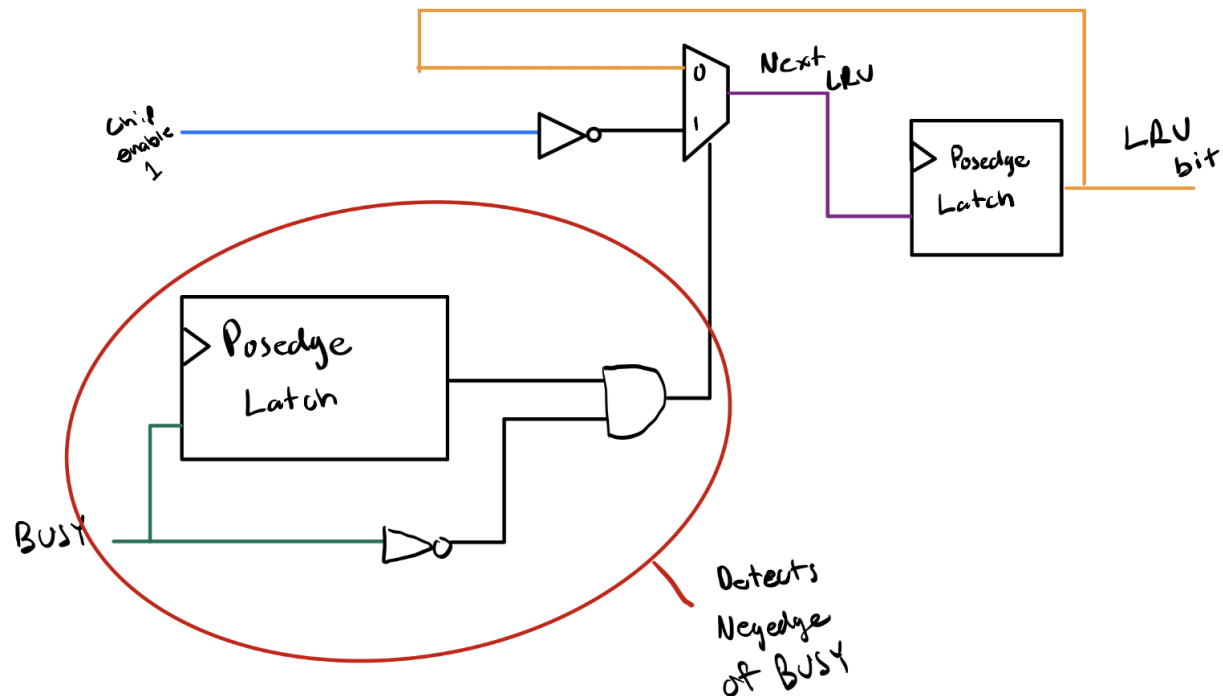


Figure 3: Updating LRU Diagram

Figure 3 shows the LRU update logic. This design makes the LRU bit update after BUSY goes low. The part circled in red is a negedge BUSY detector. The detector stores BUSY every posedge, and it is ANDed with the inverse BUSY signal. This makes it so whenever BUSY goes from HIGH to low, the signal from the latch is high and the inverse of BUSY is also HIGH. This selects the inverse chip enable 1 from the mux which will be the next LRU. Chip enable 1 corresponds to way 1 when HIGH and way 0 is low. If chip enable 1 was low that means it was just used, so the next LRU should be way 1. The next LRU is stored at every posedge.

3 2-Way Cache Functionality

3.1 Read Miss

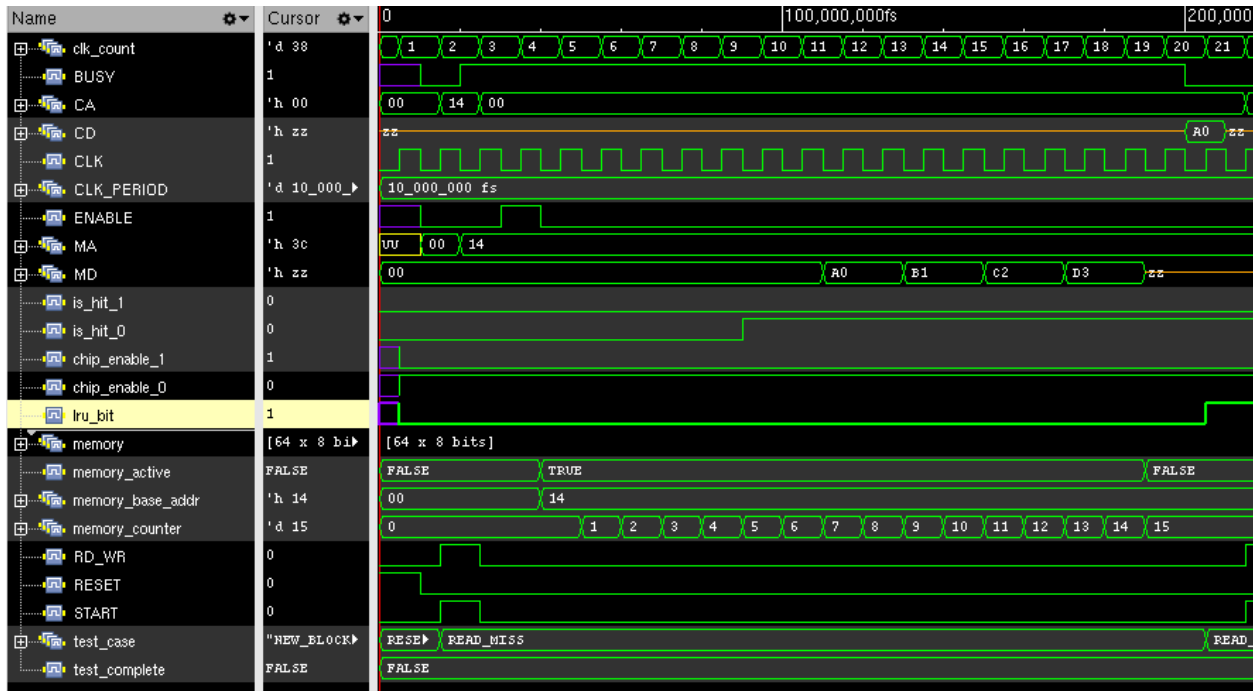


Figure 6: Read Miss Waveform

Figure 6 shows what the waveform looks like when a read miss occurs. The timings for the 2-way cache are exactly the same as the 1 way cache. The only difference is which way is selected based on the hits and misses. After the start signal, the hits from both caches are seen as low, which will make the `chip_enable_1` signal choose the LRU bit which is currently 0, meaning way 0 will be used. This makes `chip_enable_0` go high meaning the memory data will go to way 0. After the read miss operation is completed the LRU bit goes to 1 meaning way 1 is the new LRU which is correct because way 0 was just used.

3.2 Read Hit

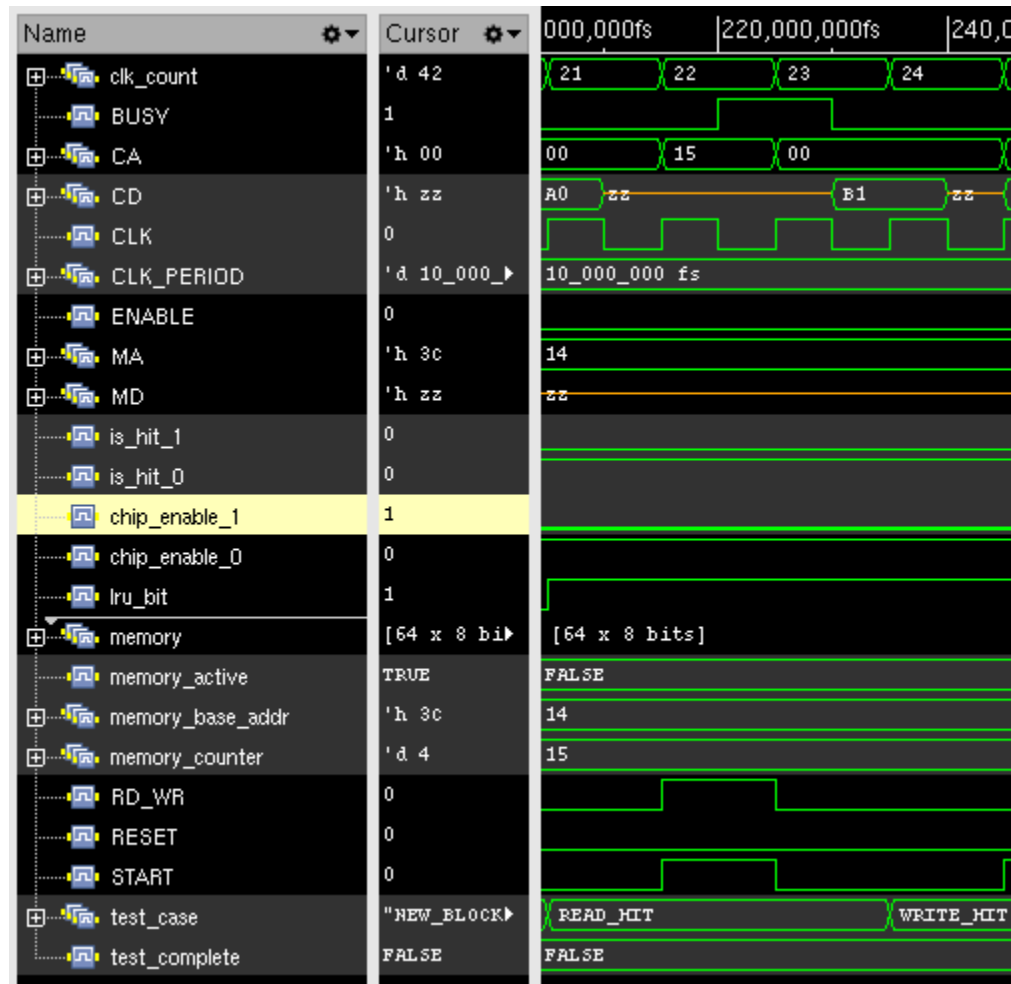


Figure 7 : Read Hit Waveform

Figure 7 shows what the waveform looks like for a read hit. The 2-way cache doesn't use the LRU bit for hits. Whichever way gets a hit, is whichever way will be enabled. Because way 0 was a hit, due to the previous read miss, chip_enable_0 stays high since that is the current way that will be read from. After the read hit finishes the LRU bit stays at 1 since way 0 was just used.

3.3 Write Hit



Figure 8: Write Hit Waveform

Figure 8 shows the waveform for a write hit. The first part of Figure 8 shows when the write hit occurs. The cache data “FF” was written to the cache address “16”. Since the hit occurred in way 0, chip_enable_0 stays high. After the write hit finishes the LRU bit stays at 1 since way 0 was just used. The second part of figure 8 shows the data being read from address “16”. This was a read hit and chip_enable_0 was HIGH because that's where the data was written to. After the check the LRU goes to HIGH because way 0 was just used.

3.4 Write Miss



Figure 9 : Write Miss Waveform

Figure 9 shows the waveform for a write miss. The first part of Figure 9 shows when a write miss occurs. When BUSY goes high both hits are 0, this means both ways are misses so the LRU bit is used to determine which cache to enable. The LRU bit is 1 at this time so way 1 will be used for the operation. After write miss finishes the LRU bit becomes 0 because way 1 was just accessed. The second part of figure 9, verifies that the write miss didn't write anything. Because a write miss didn't write anything, a read miss occurred which is correct.

4.1 Demultiplexer

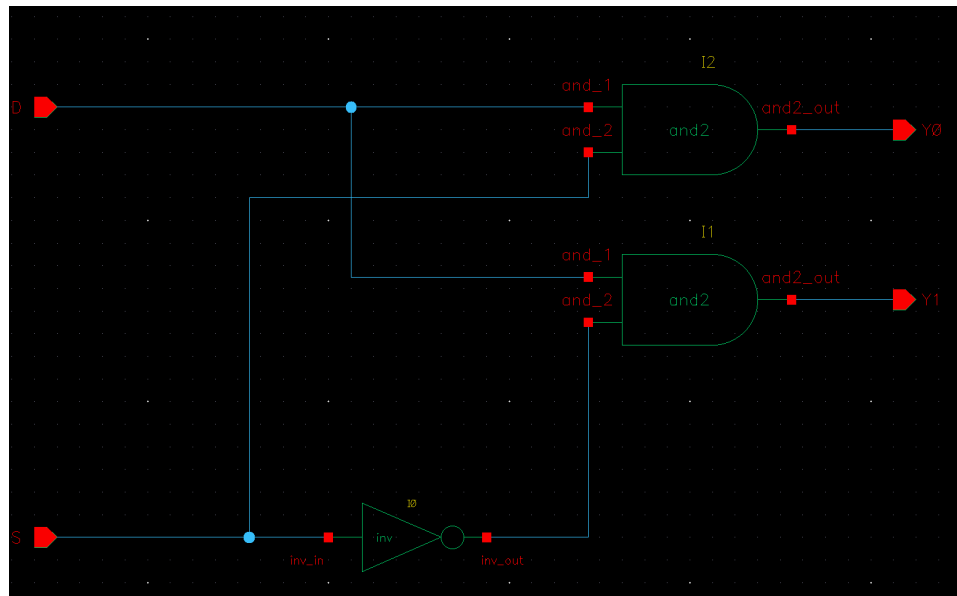


Figure 10: Demultiplexer 1 to 2 Schematic

Figure 10 shows the schematic for the 1 to 2 demultiplexer.

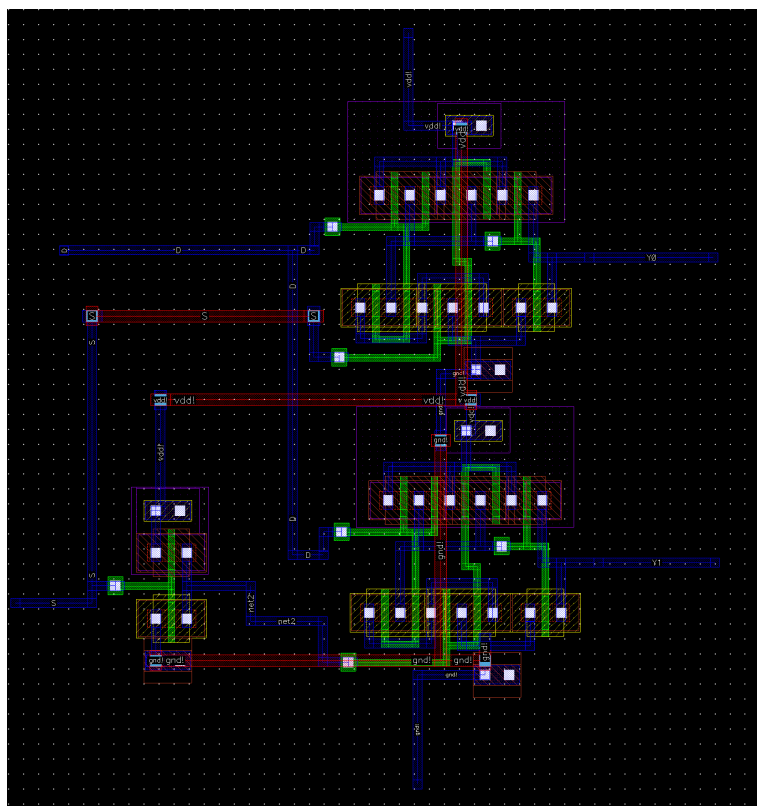


Figure 11: Demultiplexer 1 to 2 Layout

Figure 11 shows the layout for the 1 to 2 demultiplexer.

```

INFO:      Status: Finishing as of 2025-12-08 12:14:43 Elapsed: 00:00:03 CPU time[s]: 3
INFO:      Resource usage by worker at exit:
          CPU      Elapsed CPUs      CPU      Peak      Peak engine
Worker:  time[s]  time[s]  Allocated  usage[%]  Memory[MB]  Memory[MB]
          0:      2      3          1      61.02      26          12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/demux1to2.sum
INFO:      All selected rules completed
          Total CPU Time      : 2(s)
          Total Real Time     : 4(s)
          Total Original Geometry : 269(517)
          Total DRC RuleChecks  : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/demux1to2.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-12-08 12:14:43

```

Figure 12: Demultiplexer 1 to 2 DRC Test

Figure 12 shows the DRC Test for the 1 to 2 demultiplexer layout and it was able to run with 0 errors.

```

INFO:      Generating the LVS report...
          Report Init (0s)
          Load common queries (0s)
          Create report (0s)
          Cleanup (1s)
INFO:      Report finished
INFO:      Extraction Run Summary
          Total CPU Time      : 1(s)
          Total Real Time     : 2(s)
          Peak Memory Used    : 31.00(M)
INFO:      NVN Run Summary
          Total CPU Time      : 1(s)
          Total Real Time     : 1(s)
          Peak Memory Used    : 250.56(M)
INFO:      LVS Summary
          Total CPU Time      : 2(s)
          Total Real Time     : 3(s)
          Peak Memory Used    : 250.56(M)

#####
# Run Result      : MATCH
#
# Run Summary     : [INFO] ERC Results: Empty
#                 : [INFO] Extraction Clean
#
# ERC Summary File : demux1to2.sum
# Extraction Report File : demux1to2.rep
# Comparison Report File : demux1to2.rep.cls
#
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 75.45
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Dec 8 11:54:24 2025

Pegasus finished normally. 2025-12-08 11:54:24

```

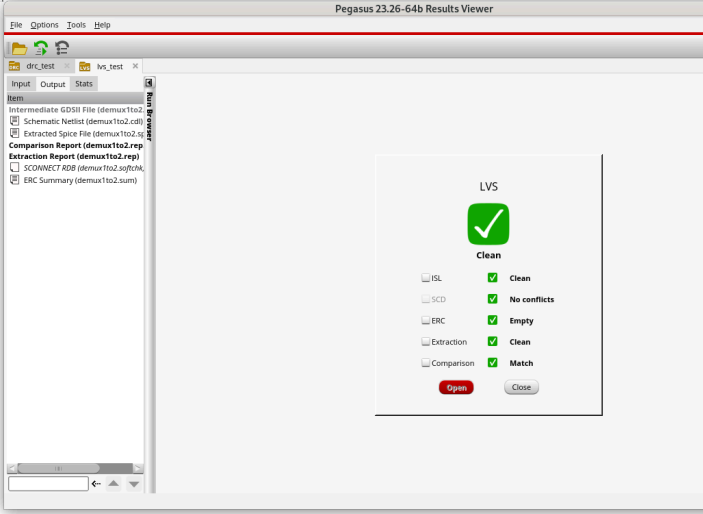


Figure 13: Demultiplexer 1 to 2 LVS Test

Figure 13 shows the LVS test for the 1 to 2 demultiplexer layout and it was able to pass.

4.2 Chip Select Logic

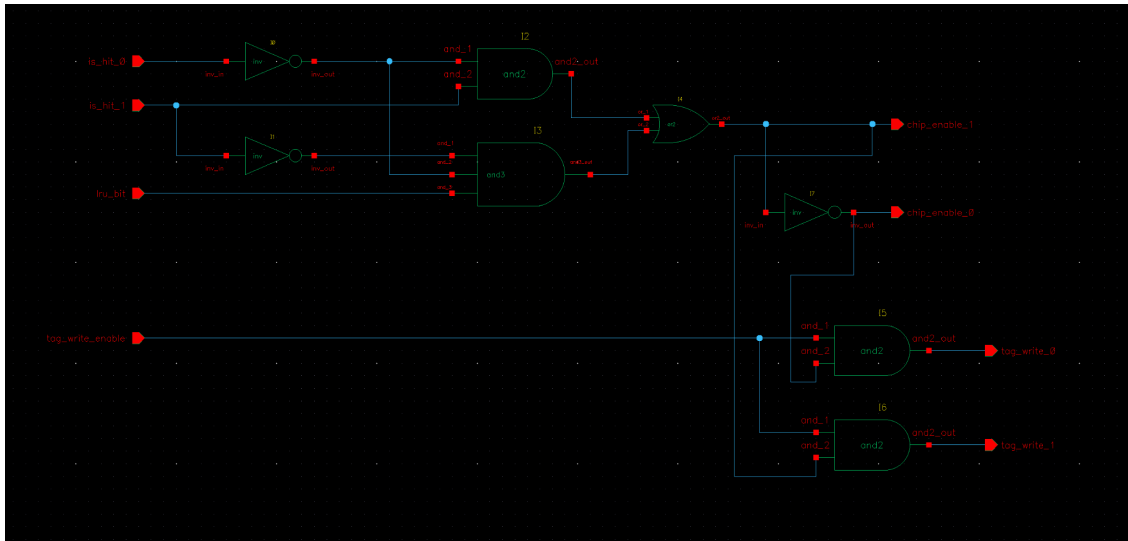


Figure 14: Chip Select Schematic

Figure 14 shows the schematic for the chip select.

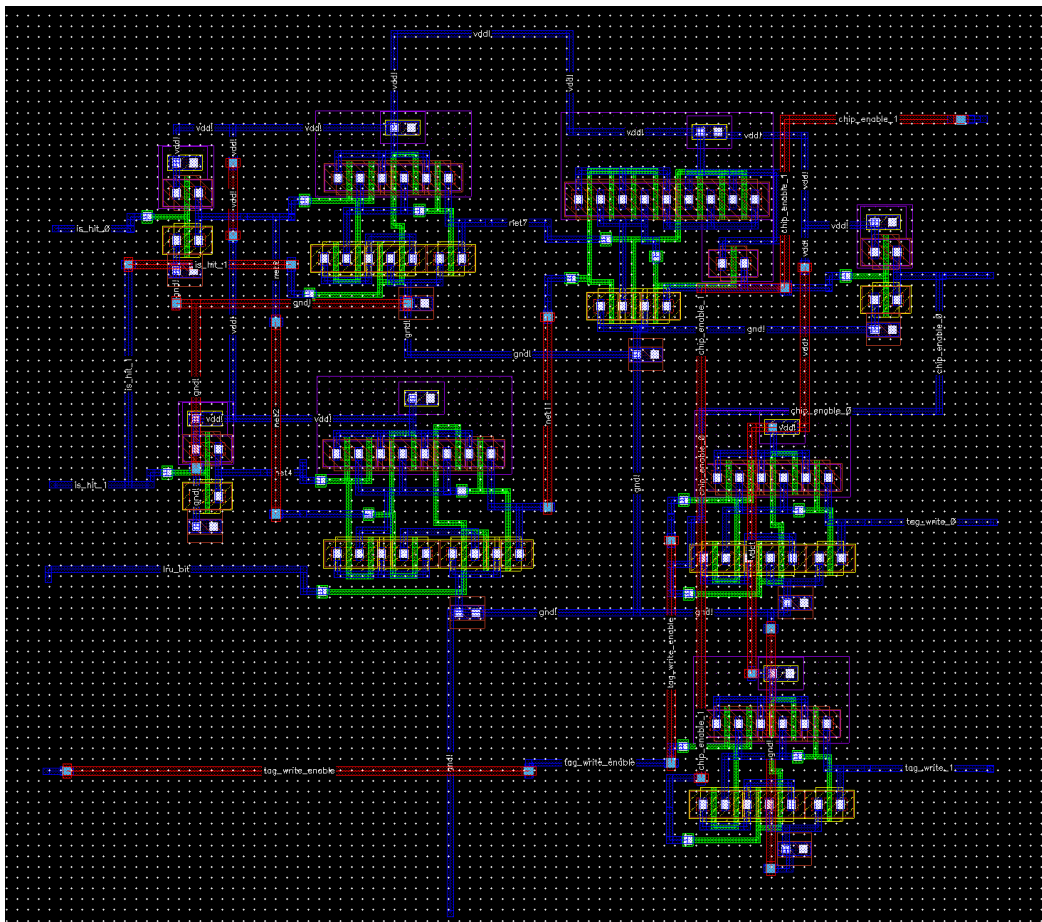


Figure 15: Chip Select Layout

Figure 15 shows the layout for the chip select.

```

INFO:      Status: Finishing as of 2025-12-08 12:36:16 Elapsed: 00:00:03 CPU time[s]: 3
INFO:      Resource usage by worker at exit:
          CPU      Elapsed CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
          0:      2      3      1      60.76      26      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/chip_select.sum
INFO:      All selected rules completed
          Total CPU Time      : 2 (s)
          Total Real Time     : 4 (s)
          Total Original Geometry : 663(1442)
          Total DRC RuleChecks  : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/chip_select.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-12-08 12:36:16

```

Figure 16: Chip Select DRC Test

Figure 16 shows the DRC Test for the chip select layout and it was able to run with 0 errors.

```

32 insts vs 32 insts . . . . . (0s)-match
INFO:      Generating the LVS report...
          Report Init (0s)
          Load common queries (0s)
          Create report (0s)
          Cleanup (0s)
INFO:      Report finished
INFO:      Extraction Run Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 2 (s)
          Peak Memory Used    : 31.00 (M)
INFO:      NVN Run Summary
          Total CPU Time      : 0 (s)
          Total Real Time     : 1 (s)
          Peak Memory Used    : 256.09 (M)
INFO:      LVS Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 3 (s)
          Peak Memory Used    : 256.09 (M)

#####
# Run Result      : MATCH
#
# Run Summary     : [INFO] ERC Results: Empty
#                  : [INFO] Extraction Clean
#
# ERC Summary File : chip_select.sum
# Extraction Report File : chip_select.rep
# Comparison Report File : chip_select.rep.cls
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 80.39
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Dec 8 12:37:26 2025

Pegasus finished normally. 2025-12-08 12:37:26

```

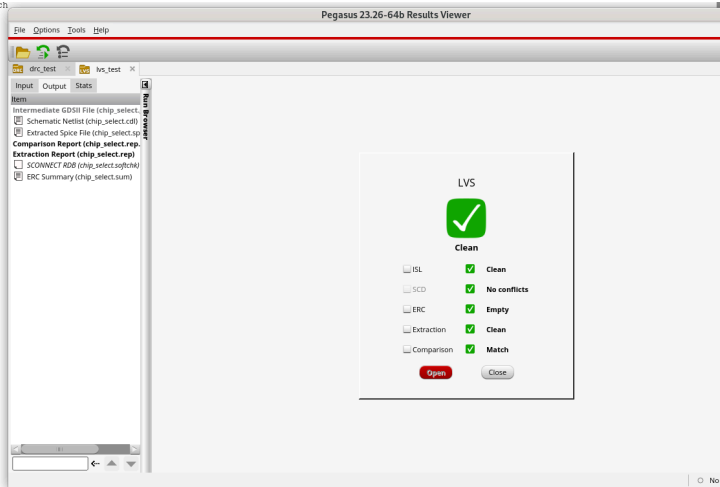


Figure 17: Chip Select LVS Test

Figure 16 shows the LVS test for the chip select layout and it was able to pass.

4.3 LRU Update Logic

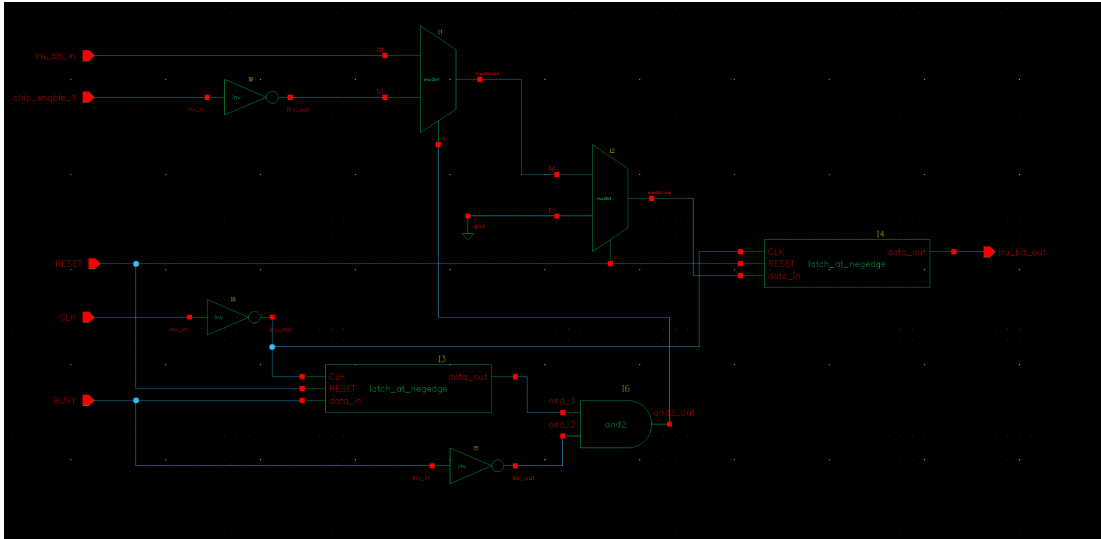


Figure 18: LRU Update Logic Schematic

Figure 18 shows the schematic for the LRU update logic.

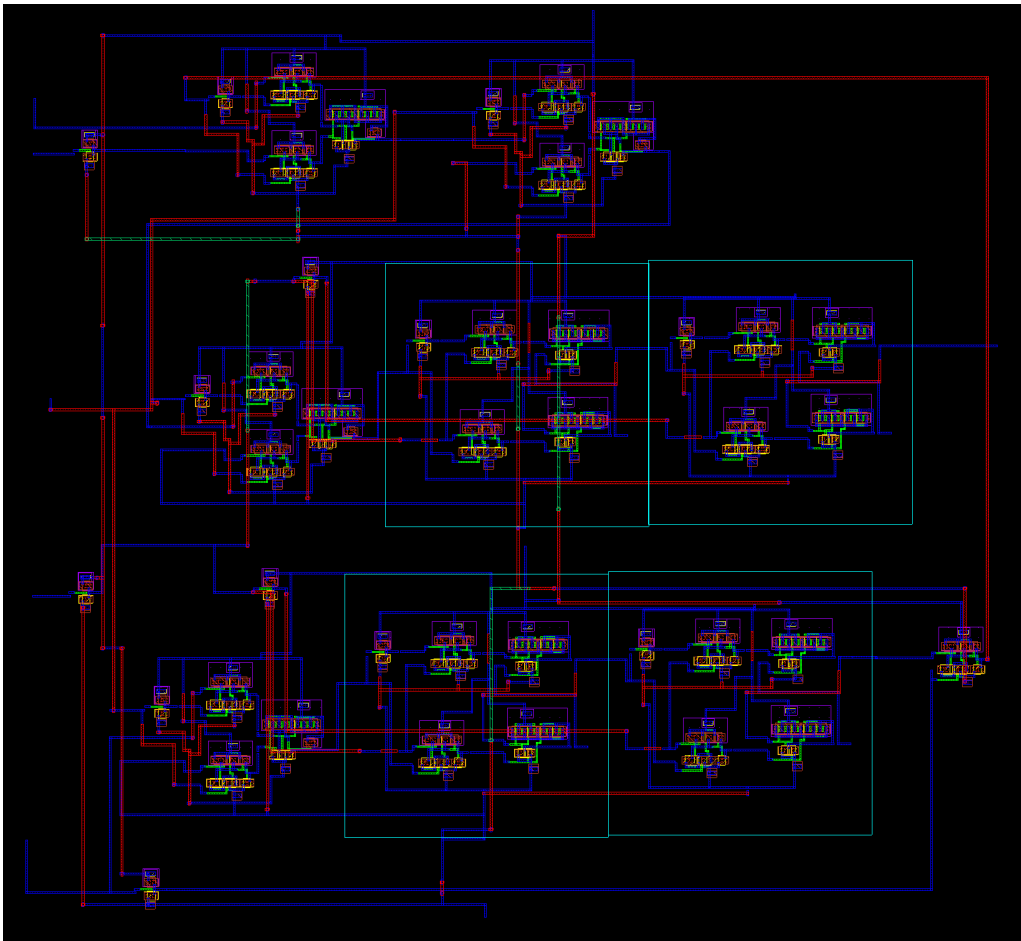


Figure 19: LRU Update Logic Layout

Figure 19 shows the layout for the LRU update logic.

```

INFO: Status: Finishing as of 2025-12-08 12:05:00 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0:      2      3      1      62.95      26      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/lru_update_logic.sum
INFO: All selected rules completed
      Total CPU Time      : 2 (s)
      Total Real Time      : 4 (s)
      Total Original Geometry : 823(7665)
      Total DRC RuleChecks    : 562
      Total DRC Results      : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/lru_update_logic.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-12-08 12:05:00

```

Figure 20: LRU Update DRC Test

Figure 20 shows the DRC Test for the LRU update logic layout and it was able to run with 0 errors.

```

155 insta vs 155 insta ..... (0s)-match
INFO: Generating the LVS report...
      Report Init (1s)
      Load common queries (0s)
      Create report (0s)
      Cleanup (0s)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time      : 1 (s)
      Total Real Time      : 2 (s)
      Peak Memory Used     : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time      : 0 (s)
      Total Real Time      : 2 (s)
      Peak Memory Used     : 258.95 (M)
INFO: LVS Summary
      Total CPU Time      : 1 (s)
      Total Real Time      : 4 (s)
      Peak Memory Used     : 258.95 (M)

#####
# Run Result      : MATCH
# Run Summary     : [INFO] ERC Results: Empty
#                 : [INFO] Extraction Clean
# ERC Summary File : lru_update_logic.sum
# Extraction Report File : lru_update_logic.rep
# Comparison Report File : lru_update_logic.rep.cls
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 81.49
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Mon Dec 8 12:13:11 2025

Pegasus finished normally. 2025-12-08 12:13:11

```

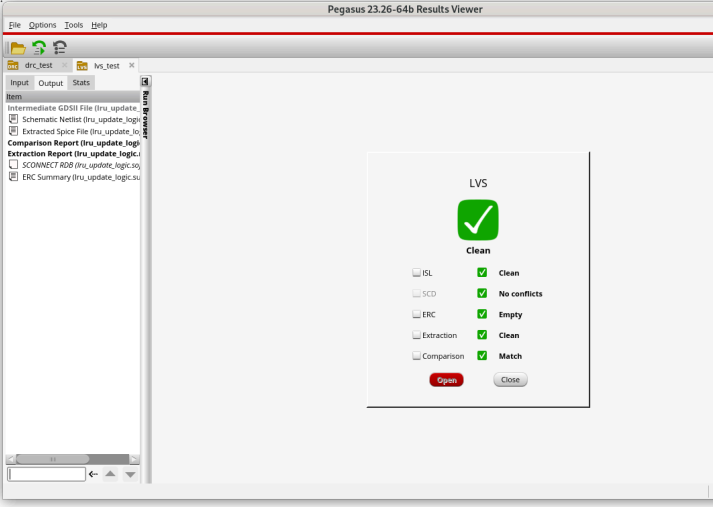


Figure 21: LRU Update LVS Test

Figure 21 shows the LVS test for the LRU update logic layout and it was able to pass.

4.4 Full 2-Way Cache

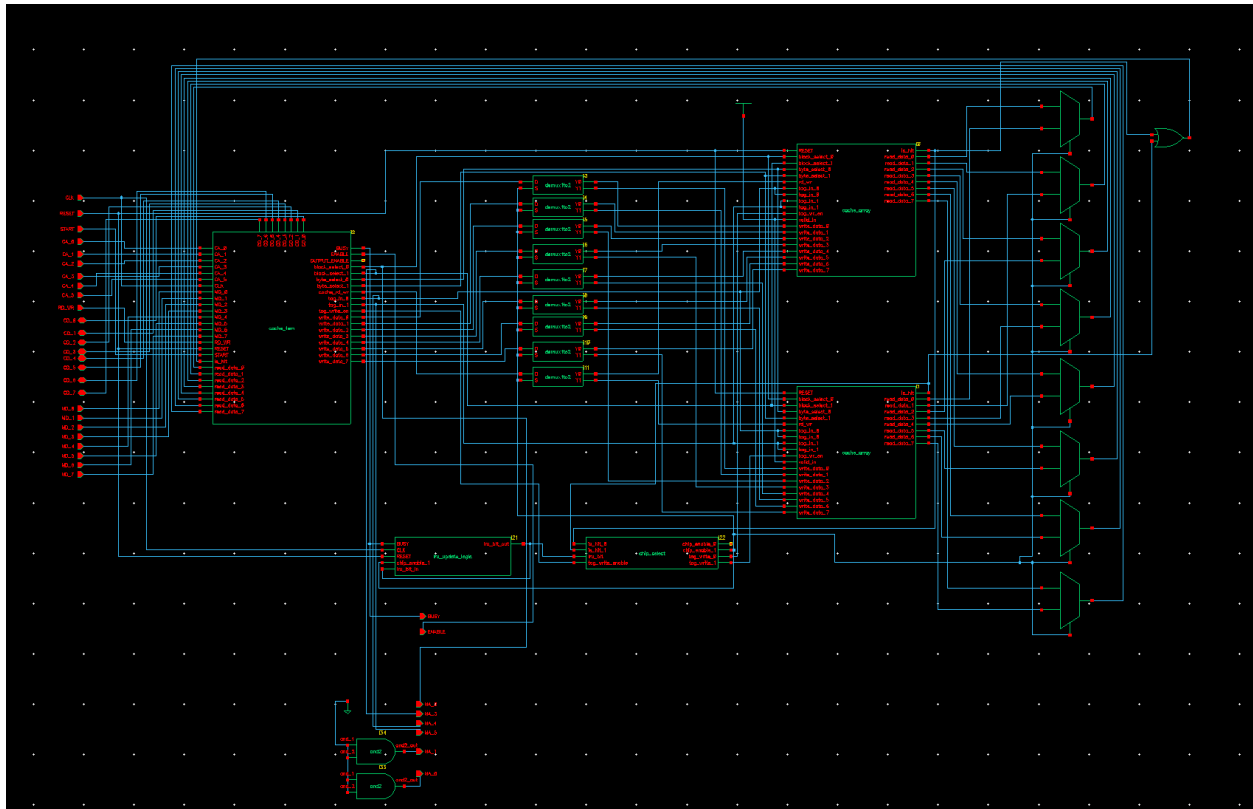


Figure 22: 2-Way Set Associative Cache Schematic

Figure 22 shows the full 2-way cache implementation schematic. The two AND gates are used to just get the memory bits 1 and 0 to be logic 0.

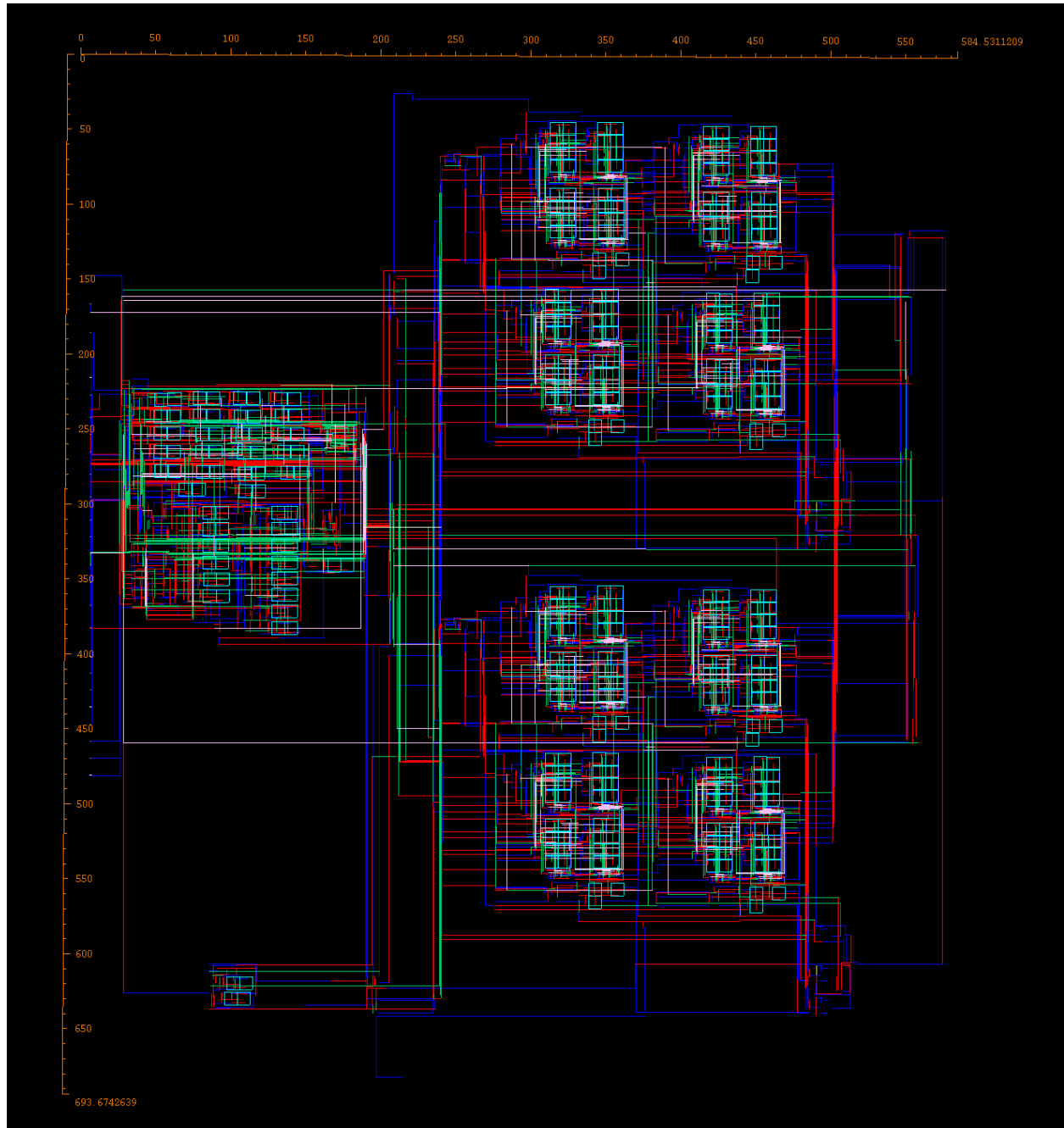


Figure 23: 2-Way Set Associative Cache Layout

Figure 23 shows the layout for the 2-way cache. Cache width 584 micro meters and height is 693 micro meters. The total area is .404 mm.

```

0:      1 / 1      121      16      3
INFO:   Status: Finishing as of 2025-12-08 15:56:42   Elapsed: 00:00:04   CPU time[s]: 4
INFO:   Resource usage by worker at exit:
        CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker:  time[s]  time[s]  Allocated  usage[%]  Memory[MB]  Memory[MB]
0:      3      4      1      70.91      121      16
Worker resource usage summary: Total CPU[s]: 3 Run duration[s]: 4 Max peak memory[MB]: 121
Worker CPU utilization summary: 3 of 4 available CPU seconds (75.00%) with 1 average available CPU(s)
INFO:   Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_2_way.sum
INFO:   All selected rules completed
        Total CPU Time      : 3(s)
        Total Real Time     : 5(s)
        Total Original Geometry : 9555(572761)
        Total DRC RuleChecks : 562
        Total DRC Results   : 0 (0)
        ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_2_way.drc_errors.ascii
INFO:   Checking in all SoftShare licenses.

Pegasus finished normally. 2025-12-08 15:56:42

```

Figure 24: 2-Way Set Associative Cache DRC Test

Figure 24 shows the DRC Test for the 2-way cache layout and it was able to run with 0 errors.

```

INFO:   Generating the LVS report...
        Report Init (0s)
        Load common queries (1s)
        Create report (0s)
        Cleanup (0s)
INFO:   Report finished
INFO:   Extraction Run Summary
        Total CPU Time      : 1(s)
        Total Real Time     : 1(s)
        Peak Memory Used    : 31.00(M)
INFO:   NVN Run Summary
        Total CPU Time      : 1(s)
        Total Real Time     : 1(s)
        Peak Memory Used    : 275.57(M)
INFO:   LVS Summary
        Total CPU Time      : 2(s)
        Total Real Time     : 3(s)
        Peak Memory Used    : 275.57(M)

#####
# Run Result      : MATCH #
# Run Summary    : [INFO] ERC Results: Empty #
#               : [INFO] Extraction Clean #
# ERC Summary File : cache_2_way.sum #
# Extraction Report File : cache_2_way.rep #
# Comparison Report File : cache_2_way.rep.cis #
#####

INFO:   Creating cross reference database ...
INFO:   Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 94.70
INFO:   Checking in all SoftShare licenses.
INFO:   Creating cross reference database completed. Mon Dec 8 16:04:22 2025

Pegasus finished normally. 2025-12-08 16:04:22

```

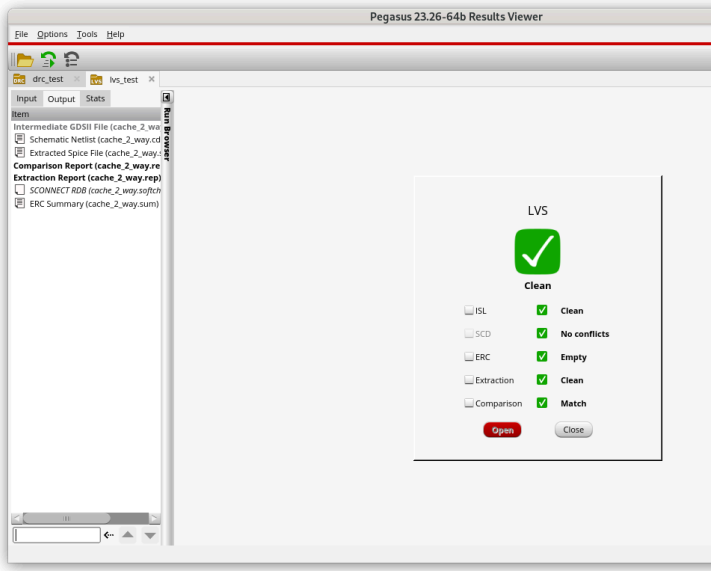


Figure 25: 2-Way Set Associative Cache LVS Test

Figure 25 shows the LVS test for the 2-way cache layout and it was able to pass.

5 Work Breakdown

Task	Person
Chip Selection VHDL	Richard
LRU Update Logic VHDL	Yomi
Full Design VHDL	Richard and Yomi
Chip Selection Layout	Yomi
LRU Update Logic Layout	Yomi
Demultiplexer Layout	Richard
Full Design Layout	Richard
Lab Report	Richard and Yomi

Note: The instructions for running the files is on the readme from the zipfile.